Neil (Hsuan Chih) Chen

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Education

New York University May 2026
Master's in **Data Science** Cumulative GPA: 3.8/4.0

University of Minnesota-Twin Cities May 2023
Bachelor of Arts in **Mathematics** Cumulative GPA: 3.6/4.0
Minor in **Statistics**

Research Experience

University of Pennsylvania — Philadelphia, PA

Researcher, Han's AI Lab June 2026 – Present
Supervisor: Dr. Tianyu Han

JEV-Agent: Explainable and Efficient Clinical Decisions from EHRs
September 2026 – Present
Manuscript in preparation

- Developed a clinical workflow for electronic health records using JEV, a model for probabilistic decisions over structured options, to select tools, acquire evidence, and update differential diagnoses.
- Implemented decision tracing through action probabilities, evolving differential diagnoses, and evidence-linked work-up logs.
- Achieved stronger overall diagnostic performance than evaluated zero-shot LLM and LLM-agent baselines, while surpassing evaluated MIRA-v2 variants on selected diseases.
- In the evaluated setup, inference cost for 546 cases was $1.31 versus $20.50–$79.70 for single-run MIRA-v2 baselines; median runtime was 2.2 versus 89.6–99.6 seconds per case.

Med-JEV: Structured Decision Learning from Medical Images
September 2026 – Present
Work in progress

- Developing a teacher–student framework that uses JEV to convert radiology reports into question-specific labels and probabilities for an image-based student.
- Designed and implemented a pipeline to generate, sample, and validate questions from existing datasets using LLMs and JEV.
- Adapting a MedGemma student with a shared scoring head to learn binary, categorical, and ordinal decisions from report-derived JEV supervision.

RL Attacker: Indirect Prompt Injection in Medical Agents
June 2026 – Present
Work in progress

- Developed a controlled framework to test indirect prompt injection in MEDDxAgent's retrieved content, using frozen retrieval and paired clean and attacked runs.
- Trained a Llama-3.1-8B attacker with GRPO and LoRA to promote a targeted incorrect diagnosis against a frozen Llama-3.1-8B victim within a single retrieval step.
- On 251 eligible held-out DDxPlus cases, targeted rank-1 flip rates were 46.6% with RL versus 31.9% with a fixed manual injection (+14.7 percentage points); single-seed evaluation with oracle-assisted target selection.

Research Experience (continued)

New York University

New York, NY

Research Assistant, Shen's Lab April 2025 – June 2026

Supervisor: Dr. Yiqiu Shen

DeepMS: Multimodal AI for Multiple Sclerosis Diagnosis

April 2025 – August 2025

Under revision for JAMA Neurology; preprint available

- Integrated age and sex into DeepMS, a multimodal MRI framework for multiple sclerosis diagnosis.
- Compared demographic–imaging fusion strategies to assess the contribution of patient metadata to diagnostic performance.

Multimodal and Longitudinal Breast Cancer Detection

September 2025 – June 2026

Manuscript in preparation

- Evaluated a multimodal transformer integrating current and prior 2D and 3D mammography, ultrasound, and clinical data for breast cancer detection.
- Curated public datasets for external validation and packaged the inference pipeline in Docker for external institutions to evaluate on their own data.
- The study reported an internal screening AUROC of 0.933 (95% CI: 0.921–0.944); external AUROCs ranged from 0.831 to 0.958 across cohorts and benchmarks using different modality inputs.
- Developed attention visualizations to examine image regions associated with model predictions across modalities.

Multimodal AI for Acute Respiratory Failure Prediction

September 2025 – January 2026

- Developed a late-fusion framework combining CheXFound imaging predictions and XGBoost clinical predictions to classify acute respiratory failure etiologies and predict mortality.
- In the reported evaluation, fusion achieved a 28-day mortality AUROC of 0.835, compared with 0.802 for imaging alone and 0.702 for clinical data alone.

Publications and Preprints

Ma, J., Stepanov, V., Rui, W., **Chen, H.-C.**, et al. (2026). Diagnosis of Multiple Sclerosis Using Multimodal Deep Learning Integrating Lesion and Normal-Appearing White Matter: A Retrospective Study with International Multicentre External Validation. *medRxiv* [Preprint]. DOI: 10.64898/2026.03.04.26347460.
Manuscript under revision for JAMA Neurology.

Manuscripts in Preparation

- JEV-Agent: Explainable and Efficient Clinical Decisions from EHRs.
- Multimodal and Longitudinal Breast Cancer Detection Using Imaging and Clinical Data.